

PAY-TV & STREAMING

STADIOchoice

Customer Churn Prediction

1. Purpose of Data Request

This data request specifies the information required from STADIOchoice to investigate and develop a data science solution for subscriber churn within STADIOstream.

The requested data is intended to support the complete analytical process, including exploratory data analysis, feature engineering, churn definition, predictive modelling, model evaluation and the interpretation of factors associated with subscriber churn.

Where possible, historical and timestamped records are requested rather than only the current state of each subscriber. This will allow subscriber behaviour to be examined before a defined prediction point and compared with subsequent churn outcomes.

All personally identifiable information that is not required for the analysis should be excluded or anonymised.

2. Subscriber and Account Information

Column name	Data type	Example	Additional information
Subscriber ID	String, identifier	SUB001245	Unique anonymised identifier for each subscriber. The identifier must remain consistent across all supplied datasets.
Household ID	String, identifier	HH000892	Identifier for the household associated with the subscription, where applicable.
Account creation date	Datetime	2025-01-12 10:42:18	Date and time at which the customer account was created.
Acquisition date	Datetime	2025-01-12 10:45:02	Date and time at which the subscriber started the service.
Acquisition channel	String, categorical	Digital	Channel through which the subscriber was acquired. Valid values and definitions should be supplied by the client.

Acquisition campaign	String, categorical	Sports Campaign	Campaign or promotion associated with the subscriber acquisition, where applicable.
Customer segment	String, categorical	Premium	STADIOchoice-defined customer segment, where applicable.
Province	String, categorical	Gauteng	South African province, where this information is legitimately collected and available.

3.Subscription Information

Column name	Data type	Example	Additional information
Subscriber ID	String, identifier	SUB001245	Must match the subscriber identifier in other datasets.
Subscription ID	String, identifier	SUBS009832	Unique identifier for the subscription.
Subscription plan	String, categorical	Premium Streaming	Exact STADIOstream package subscribed to.
Subscription start date	Datetime	2025-01-12 10:45:02	Date and time at which the subscription became active.
Subscription end date	Datetime	2025-08-31 23:59:59	Date and time at which the subscription ended. Null where the subscription remains active.
Subscription status	String, categorical	Active	Valid values and definitions should be documented.
Monthly fee	Numeric, decimal	199.00	Monthly subscription charge. Currency must be specified.
Discount amount	Numeric, decimal	50.00	Discount applied to the subscription for the relevant period.
Promotion code	String, categorical	SPORT50	Promotion associated with the subscription, where applicable.

4. Viewing Behaviour

The viewing behaviour data is intended to capture how subscribers interact with STADIOstream content over time.

Column name	Data type	Example	Additional information
Subscriber ID	String, identifier	SUB001245	Must match the subscriber identifier used across datasets.
Session ID	String, identifier	SES827361	Unique identifier for a viewing session.
Viewing timestamp	Datetime	2026-05-14 20:31:45	Timestamp of the viewing event. A consistent timezone must be used.
Content ID	String, identifier	CNT004582	Unique identifier for the content accessed.
Content category	String, categorical	Live Sport	Category or genre of the content. Valid categories should be documented.
Content duration	Numeric, integer	7200	Total content duration, preferably recorded in seconds.
Viewing duration	Numeric, integer	3840	Actual viewing duration, preferably recorded in seconds.
Completion rate	Numeric, decimal	0.53	Proportion of content watched. Valid range: 0–1.
Device type	String, categorical	Smart TV	Device used to access the content.
Pause count	Integer	3	Number of pauses during the viewing session.
Abandonment indicator	Boolean	True	Indicates whether the subscriber abandoned the content before completion.

5. Engagement Behaviour

Column name	Data type	Example	Additional information
Subscriber ID	String, identifier	SUB001245	Must match the subscriber identifier used across datasets.
Event timestamp	Datetime	2026-05-14 20:25:12	Date and time of the interaction.
Event type	String, categorical	Search	Examples may include Search, Play, Pause, Scroll and other tracked events. The complete set of valid values should be provided.
Search term	String, text	Champions League	Search query entered by the subscriber, where applicable.
Search result count	Integer	12	Number of results returned by the search.
Content click indicator	Boolean	True	Indicates whether the subscriber selected a content result.
Scroll count	Integer	8	Number of scrolling interactions during the relevant interaction or session.

6. Billing and Payment History

Column name	Data type	Example	Additional information
Subscriber ID	String, identifier	SUB001245	Must match the subscriber identifier used across datasets.
Billing date	Datetime	2026-06-01 00:00:00	Date and time of the billing event.
Amount charged	Numeric, decimal	199.00	Amount charged for the billing period. Currency must be specified.
Payment status	String, categorical	Paid	Valid values should include documented statuses such as

			Paid, Failed, Pending or Refunded.
Payment method	String, categorical	Debit Order	Method used to make payment.
Payment failure indicator	Boolean	False	Indicates whether the payment attempt failed.
Payment failure count	Integer	2	Number of failed payment attempts within the relevant period.
Refund indicator	Boolean	False	Indicates whether a refund occurred.
Refund amount	Numeric, decimal	0.00	Amount refunded, where applicable.

7. Cancellation and Retention History

Column name	Data type	Example	Additional information
Subscriber ID	String, identifier	SUB001245	Must match the subscriber identifier used across datasets.
Cancellation date	Datetime	2026-06-30 18:42:10	Date and time the subscriber cancelled. Null where no cancellation occurred.
Cancellation reason	String, categorical	Price	Reason recorded for cancellation. Valid categories and definitions should be supplied.
Cancellation channel	String, categorical	Mobile App	Channel through which the cancellation occurred.
Churn indicator	Boolean	True	Indicates whether the subscriber meets STADIOchoice's official churn definition.
Reactivation indicator	Boolean	False	Indicates whether the subscriber subsequently returned to the service.

Reactivation date	Datetime	2026-08-15 12:00:00	Date of reactivation, where applicable.
Retention interaction date	Datetime	2026-06-25 13:12:44	Date and time of a retention interaction.
Retention offer	String, categorical	20% Discount	Retention offer presented to the subscriber, where applicable.
Offer accepted	Boolean	True	Indicates whether the subscriber accepted the retention offer.
Offer outcome	String, categorical	Retained	Outcome following the retention intervention.

8. Data Definitions and Metadata

In addition to the datasets above, the following documentation is requested to ensure that the supplied data can be correctly interpreted, validated and prepared for analysis.

Requirement	Data type	Example	Additional information
Churn definition	Text	Subscription cancelled and not reactivated within 30 days	Provide STADIOchoice's official business definition of churn, including how temporary suspensions, failed payments and reactivations are treated.
Observation period	Date range	2024-01-01 to 2026-06-30	Specify the complete historical period covered by the supplied data.
Data extraction date	Datetime	2026-09-01 08:00:00	Date and time at which the data extract was generated.
Timezone	String	Africa/Johannesburg	All timestamps should use a documented and consistent timezone.
Missing-value definition	Text	NULL	Document how missing, unknown and not-applicable values are represented.
Categorical definitions	Text	Active = currently subscribed	Provide valid values and definitions for categorical fields.

Data dictionary	Document	Data Dictionary v1.0	Provide descriptions, units and business meanings for all supplied fields.
Key relationships	Text	Subscriber ID → Subscription ID	Document primary keys, foreign keys and relationships between datasets.
Known data issues	Text	Viewing events missing for 2% of subscribers	Document known data-quality issues, historical system changes, coverage limitations or other relevant constraints.

9. Additional Data Requirements

The following requirements apply across the requested datasets:

- Historical records should be supplied wherever available rather than only the latest status of each subscriber.
- Each dataset should contain a consistent anonymised Subscriber ID to enable records to be joined.
- Event-level information should include timestamps wherever possible so that customer behaviour can be analysed chronologically.
- Dates and timestamps should follow a consistent format and timezone.
- Categorical variables should use consistent values and should be accompanied by definitions of valid categories.
- Missing values should be represented consistently and documented.
- Numerical variables should specify their units, particularly monetary values and time-based measures.
- The supplied data should include both subscribers who churned and subscribers who remained active during the observation period.
- The client should provide the official business definition of churn and explain how cancellations, suspensions, failed payments and subsequent reactivations are treated.
- Personally identifiable information such as names, identity numbers, full addresses and payment-card details is not required for this project and should be excluded or anonymised.
- The client should identify known data-quality issues, structural changes to source systems and periods where data collection may have been incomplete.